

LOCH GLASCARNOCH, United Kingdom

WMO: 030310

Lat: 57.7250N

Lon: 4.8956W

Elev: 264

StdP: 98.19

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-6.2	-10.0	1.6	-8.3	-7.2	2.1	-4.8	18.7	4.9	16.4	4.4	0.6	270	0.735

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	21.7	15.8	19.6	14.7	17.9	13.8	16.7	20.3	15.6	18.6	14.5	17.1	4.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	11.1	18.1	14.1	10.4	17.0	13.2	9.7	16.0	47.8	20.3	44.4	18.7	41.5	17.2	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	14.3	12.6	11.1	DB	-11.3	24.8	3.0	1.9	-13.5	26.2	-15.3	27.3	-17.0	28.3	-19.1	29.7
(5)				WB	-11.2	18.9	3.0	1.8	-13.3	20.2	-15.1	21.2	-16.7	22.2	-18.9	23.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	2.1	2.4	3.5	5.9	8.2	10.9	12.8	12.3	10.7	7.6	4.4	2.2
	DBStd	4.94	3.46	3.34	3.13	2.93	3.07	2.60	2.43	2.31	2.46	2.79	3.34	4.19
	HDD10.0	1397	244	214	201	128	72	19	2	4	21	82	168	242
	HDD18.3	4156	502	447	459	374	313	222	172	187	231	332	417	500
	CDD10.0	284	0	0	0	3	18	47	89	77	40	8	1	1
	CDD18.3	1	0	0	0	0	0	0	1	1	0	0	0	0
	CDH23.3	19	0	0	0	0	3	5	7	4	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.6	5.1	5.3	5.4	4.6	4.3	4.3	3.9	4.0	4.4	4.6	4.5	4.8
Precipitation	PrecAvg	1566	199	148	130	97	98	78	79	97	115	166	184	182
	PrecMax	1941	363	308	330	164	211	117	130	236	222	289	314	347
	PrecMin	1020	15	20	20	45	13	36	36	38	39	51	66	4
	PrecStd	243	101	81	63	30	47	22	29	47	51	67	66	99
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	11.7	14.0	18.1	22.7	23.6	24.2	23.5	20.7	15.9	13.3	12.2
		MCWB	9.2	9.3	8.4	11.5	15.1	16.1	17.4	17.4	15.2	12.5	11.2	10.2
	2%	DB	9.6	10.3	10.8	14.7	19.1	19.9	21.7	19.8	18.0	14.2	11.7	10.8
		MCWB	8.0	8.3	8.1	9.9	12.6	14.3	16.4	15.5	14.1	11.9	9.9	9.2
	5%	DB	8.4	9.0	9.5	12.7	16.3	17.6	19.4	17.9	16.2	13.1	10.6	9.7
		MCWB	6.8	7.2	7.4	8.9	11.2	13.1	15.3	14.4	13.3	11.1	8.8	8.2
	10%	DB	7.1	7.4	8.4	11.0	14.0	15.7	17.5	16.5	14.9	12.0	9.4	8.2
		MCWB	5.7	5.8	6.5	8.0	10.1	12.2	14.2	13.6	12.4	10.0	7.9	6.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	10.0	9.7	12.1	15.9	17.0	18.5	18.2	16.2	13.5	11.6	10.7
		MCDB	10.8	11.3	12.0	15.8	21.8	22.0	22.4	21.2	18.6	15.1	13.0	11.9
	2%	WB	8.2	8.7	8.5	10.5	13.3	15.1	16.9	16.2	14.7	12.3	10.3	9.4
		MCDB	9.4	10.2	10.4	14.1	17.8	19.2	20.6	19.0	17.0	13.8	11.6	10.7
	5%	WB	7.0	7.5	7.7	9.4	11.8	13.7	15.7	15.0	13.7	11.3	9.0	8.3
		MCDB	8.4	8.9	9.4	12.2	15.7	16.7	18.8	17.2	15.8	12.8	10.2	9.6
	10%	WB	5.8	6.0	6.7	8.3	10.4	12.6	14.6	14.0	12.7	10.3	8.0	6.8
		MCDB	7.1	7.3	8.3	10.6	13.4	15.2	17.2	16.0	14.7	11.8	9.3	8.2
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.9	6.4	7.7	8.7	7.6	7.4	7.2	7.3	6.4	5.7	5.8
		MCDBR	6.0	6.9	8.3	11.3	13.6	11.3	11.3	10.1	9.7	7.5	6.6	6.8
		MCWBR	5.1	5.6	6.1	7.1	8.0	6.6	6.7	6.4	6.5	5.8	5.7	6.3
	5% WB	MCDBR	5.8	6.2	7.5	9.9	12.3	10.3	10.1	9.1	8.8	6.9	6.5	6.7
		MCWBR	5.2	5.5	5.9	6.6	7.6	6.5	6.4	6.4	6.4	5.7	5.9	6.3
Clear-Sky Solar Irradiance	taub	0.318	0.336	0.356	0.372	0.374	0.367	0.369	0.356	0.347	0.333	0.313	0.304	
	taud	2.157	2.253	2.282	2.291	2.336	2.390	2.397	2.439	2.458	2.394	2.228	2.101	
	Ebn at Noon	506	689	786	836	860	871	860	848	798	695	514	395	
	Edh at Noon	54	76	96	111	114	109	106	95	81	65	49	42	
All-Sky Solar Radiation	RadAvg	0.36	0.96	1.98	3.37	4.54	4.43	4.16	3.33	2.30	1.20	0.49	0.25	
	RadStd	0.05	0.12	0.17	0.25	0.35	0.43	0.40	0.21	0.17	0.13	0.05	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page